Caleb Rotello

Quantum Information Researcher || Writer || Eagle Scout

Github || LinkedIn || caleb@rotello.dev || My Website || GoogleScholar

Education

Ph.D. in **Electrical and Computer Engineering** at **University of California, Los Angeles** *Los Angeles, CA*
Advisor: Leo Zhou Sep 2025 –

M.S. in **Quantum Engineering** at **Colorado School of Mines** *Golden, CO*
Thesis: *Symmetry-protected subspaces in quantum simulations* May 2021 – Aug 2022
Advisor: Eliot Kapit

B.S. in **Computer Science** at **Colorado School of Mines** *Golden, CO*
GPA 3.79/4.0 (Major GPA 4.0); Magna Cum Laude Aug 2018 – May 2021

Research Experience

- **Graduate Student** at **UCLA** Sep 2025 – Present
  Mentored by Professor Leo Zhou, *Electrical and Computer Engineering*
  - Developed mathematical tools to analyze quantum optimization methods
- **Researcher II** at **NREL** July 2023 – May 2025
  Complex Systems, Simulation, and Optimization Group
  - Developed quantum computer algorithms for stochastic programming, computing *m*-point chemical correlation functions, and optimization.
  - Led research and software development for quantum computing resource estimation and benchmarking on real-world energy problems.
  - Proposal development for: integrated systems of quantum sensors, networks, and computers on the power grid; practical quantum computing; quantum algorithms for stochastic programming.
  - Mentored interns Jacob (Coby) Sagal (now PhD student at Virginia Tech, Physics) and Mandy Bowman.
- **Machine Learning Research Intern** at **NREL** Nov 2021 – July 2023
  Mentored by Dr. Eric B. Jones and Dr. Peter Graf
  - Derived analytical methods using graph theory to predict symmetries in quantum simulations.
  - Researched the application of quantum computers to the power grid.
- **Graduate Research Assistant** at **CSM** Summer 2021 – Summer 2022
  Mentored by Professor Eliot Kapit, *Physics*
  - Used Google Sycamore processor to calculate spin-foam amplitudes in loop quantum gravity.
  - Created software to analyze different quantum algorithms for boolean SAT problems.
  - Mentored undergraduate physics student Jacob Millar (now engineer at Salesforce).
- **Bioinformatics Research Intern** at **Invitae** (prev ArcherDX) Summer 2020, 2021
  Mentored by Abel Licon
  - Developed linear time algorithm with Levenshtein automata to pair noisy DNA sequences.
  - Wrote algorithms to pair RNA alpha and beta sequences across multiple cells.
- **Undergraduate Researcher** at **CSM**
  Mentored by Professor Judith Klein, *Chemistry* Fall 2020
  - Created sentiment analysis models to predict Covid-19 outbreaks from Twitter data.
  Mentored by Professor Tom Williams, *Computer Science* Spring, Summer 2020
  - Wrote software to recognize deictic gestures (gestures that convey meaning).

Publications & Manuscripts

- **C. Rotello**. *Quantum algorithm for approximating the expected value of a random-exist quantified oracle*. arXiv:2412.00567 (2024). (in review, epj Quantum Information).

- E. B. Jones, …, **C. Rotello**, …, W. Jones. *Dynamic, Symmetry-Preserving, and Hardware-Adaptable Circuits for Quantum Computing Many-Body States and Correlators of the Anderson Impurity Model*. Phys. Rev. Research 7, 023186 (2025).
- **C. Rotello**, …, W. Jones. *Calculating the expected value function of a two-stage stochastic optimization program with a quantum algorithm*. arXiv:2402:15029 (2024). (in review, Quantum)
- **C. Rotello**, …, E. Kapit. *Automated detection of symmetry-protected subspaces in quantum simulations*. Physical Review Research, 5, 033082 (2023).
- **C. G. Rotello**. *Symmetry Protected Subspaces in Quantum Simulations*. MS Thesis, Colorado School of Mines, (2022).

Presentations

- *Benchmarks and Resource Estimation of a Quantum Algorithm for Unit Commitment*
  - Poster at American Physical Society March Meeting 03.2025
- *Quantum algorithms for calculating the expected value of #P-Hard stochastic programming problems*
  - Poster at Southwest Quantum Information and Technologies workshop 10.2024
- *Solving randomly-quantified QUBO on a quantum processor*
  - Invited talk at Quantinuum 09.2024
- *Automated detection of symmetry-protected subspaces in quantum simulations*
  - Accepted talk at Southwest Quantum Information and Technologies workshop 11.2023
  - Invited talk at Superconducting Quantum Materials and Systems Center, Fermilab 08.2022

Honors & Awards

- **Dean's Scholar Award** at University of California-Los Angeles 2025
- **Key Contributor Award** at National Renewable Energy Laboratory 2024
- **1st Place** at IBM Humans vs. Quantum Computers Challenge in Hackathon by Womanium 2022
- **Dean's List** at CSM 2018-21
- **2nd Place** at CS Algobowl Hackathon by CSM Computer Science Department 2020
- **4th Place** HackCU VI International Hackathon 2020
- C-MAPP Transamerica Scholar by CSM Computer Science Department 2019-20
- NASA RASC-AL **Semifinalist** for Gateway-based Cis-lunar Tug 2019

Leadership & Service

- Reviewer, Quantum Information Processing Conference 2025
- Undergraduate Mentor, Quantum Computing Student Association 2025
- Advisory Committee, Mountain Vista High School Engineering Program 2024 –
- Study Group, **DARPA ISAT**, "CATS: Countering Adversarial Technology Systems" 2024 – 25
- Vice President, CSM Creative Writing Club 2021 – 22
- Poetry Editor, High Grade Journal for Creative Arts 2021 – 22
- **Eagle Scout**, Boy Scouts of America (awarded '17) 2011 – 17

Work Experience

Research Scientist	NREL	2023 – 2025
Research Intern	NREL	2021 – 2023
Research Assistant	Department of Physics, CSM	2021 – 2022
Bioinformatics Intern	Invitae	2020 – 2021
Tutor	Contracting	2019 – 2020
App Developer	Contracting with Police Depts.	2019
IT Technologist	Valor Christian High School	2017 – 2018

Skills

- Quantum information science, computational chemistry, optimization, machine learning, software dev.
- Organic chemistry, photolithography, clean room, evaporation, low temperature measurement